

Name(s): _____

DATA 101 Assignment 13: Image Analysis (BONUS)

Work in teams of 2

Instructions. Create a new Jupyter Notebook. Use Markdown cells for all text explanations/headings and Code cells for Python code. Once finished, export the notebook to HTML via **File** → **Save and export notebook as** → **HTML (.html)** and bring a printed copy to class.

1. Go to https://en.wikipedia.org/wiki/Wikipedia:Featured_pictures/Artwork/Paintings and pick a painting of your choice. Download it into your computer as .jpeg, .jpg, or .png file. These are under public domain so downloading and analyzing them is not an issue.
 - (a) Read about the painting's history. Who was the artist? When did they draw it? From which era is it considered to be from?
 - (b) From visually inspecting the painting, what do you think about the choices of color, color distribution, and brightness?
 - (c) Read the image into python using the open-cv library. Make sure to `pip install opencv-python` to install open-cv in your computer.
 - (d) Create a color distribution graph that includes a line for each color: red, blue, green. Make sure to have pixel color intensity on the x axis and count of pixels on the y axis. Interpret your graph.
 - (e) Find 5 color groups that summarize the colors in the painting using the k-means algorithm. Show the centroid pixel color of each color group. Interpret your findings.
 - (f) Create a brightness distribution histogram. Interpret your graph.